



Gosford High School

Student Number

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2017

Preliminary Final Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 8 – 11 show relevant mathematical reasoning and/or calculations

Total Marks – 55

Section I Questions 1 – 7 **7 marks**

Allow about 10 minutes for this section

Section II Questions 8 – 11 **48 marks**

Allow about 1 hour and 20 minutes for

Section 1

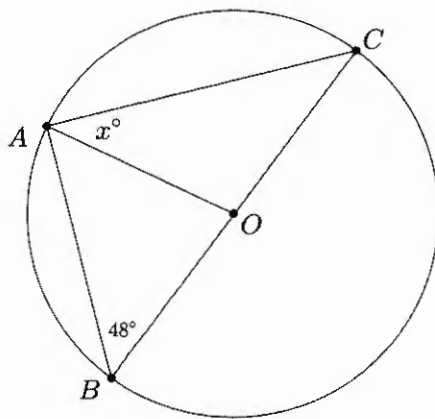
7 marks

Attempt Questions 1 – 7

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 7.

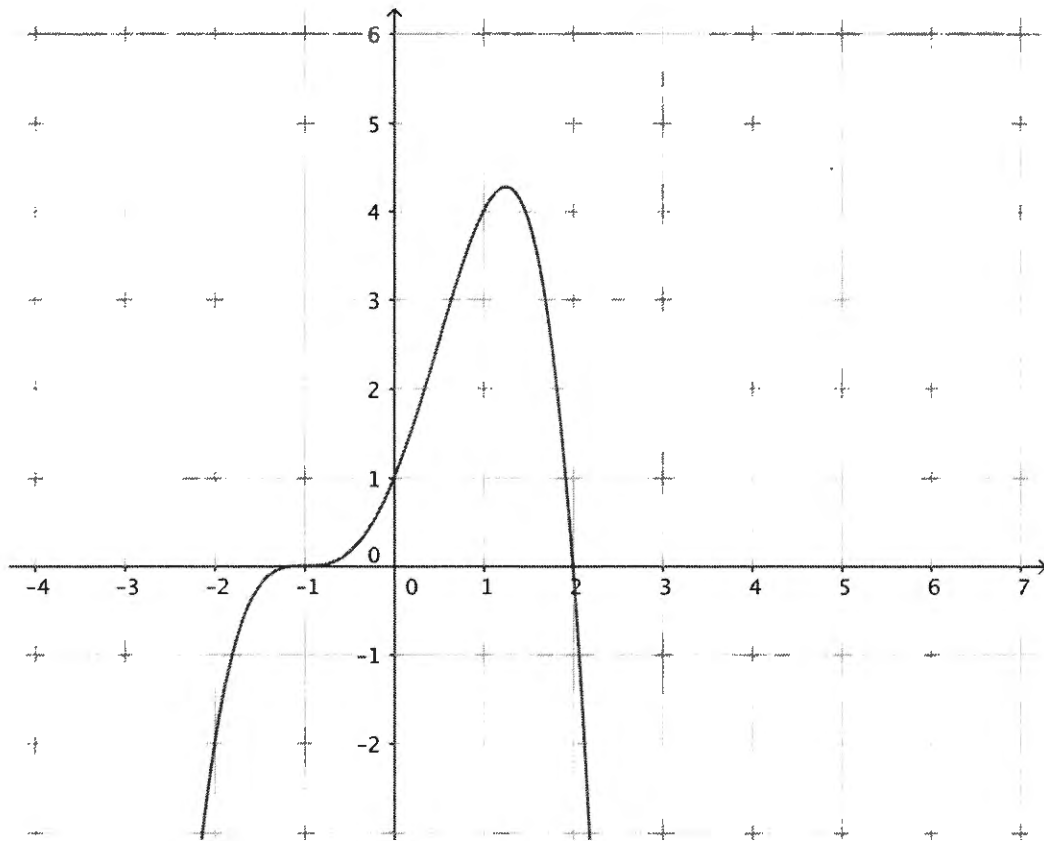
- 1 The diagram below shows a circle with centre O . Points A , B and C lie on the circle and $\angle OBA = 48^\circ$.



If $\angle OAC = x^\circ$, which of the following is the value of x ?

- (A) 48
- (B) 42
- (C) 45
- (D) 52

- 2 Which of the following is equivalent to $\sqrt{3}\sin\theta - \cos\theta$?
- (A) $2\sin(\theta + 30)^\circ$
- (B) $2\sin(\theta - 30)^\circ$
- (C) $2\sin(\theta + 60)^\circ$
- (D) $2\sin(\theta - 60)^\circ$
- 3 The number of different arrangements of the letters of the word *SERVICES* which begin and end with the letter *S* is:
- (A) $\frac{6!}{(2!)^2}$
- (B) $\frac{8!}{(2!)^2}$
- (C) $\frac{6!}{2!}$
- (D) $\frac{8!}{2!}$



Which of the following could be the equation of the graph shown above?

(A) $y = \frac{1}{2}(x+1)^3(x-2)$

(B) $y = \frac{1}{2}(x+1)^3(2-x)$

(C) $y = (x+1)^3(x-2)$

(D) $y = (x+1)^3(2-x)$

- 5 What is the solution to $|2x-1| \geq |x+1|$?
- (A) $0 \leq x \leq 3$
- (B) $x \leq 0$ or $x \geq 3$
- (C) $0 \leq x \leq 2$
- (D) $x \leq 0$ or $x \geq 2$
- 6 If $\sin x = \frac{3}{5}$, $90^\circ \leq x \leq 180^\circ$, evaluate $\tan 2x$.
- (A) $-\frac{7}{24}$
- (B) $-\frac{24}{7}$
- (C) $\frac{7}{24}$
- (D) $\frac{24}{7}$
- 7 A root of the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$ is
- (A) 2
- (B) -3
- (C) 1
- (D) -4

Section II

48 marks

Attempt Questions 8 – 11

Allow about 1 hour and 20 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 8 – 11, your responses should include relevant mathematical reasoning and/or calculations.

Question 8 (12 marks) Use a SEPARATE writing booklet.

- (a) The monic polynomial $P(x)$ has constant term 6 and is such that $P(x) = (x+3)(x-2)(ax+b)$. Find the values of a and b . 2
- (b) The lines $y = 2mx + 1$ and $y = mx + 7$ intersect at an acute angle whose tangent is $\frac{1}{3}$. Show that the possible values of m are $m = \frac{1}{2}, 1$. 3
- (c) Solve the inequality $\frac{x+3}{2x} > 1$. 2
- (d) The staff in an office consists of 6 males and 5 females. How many committees of 5 staff can be chosen which contain exactly 2 females? 2
- (e) Solve the equation $\sin 2x = \tan x$ for $0^\circ \leq x \leq 180^\circ$. 3

Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^3 - 2x^2 + kx + c = 0$ has roots $x = -1, x = 1$ and $x = 2$. 2

Show that the value of k is -1 .

- (b) If $P(3, 4)$ divides AB internally in the ratio $2:3$ and B is $(-2, 1)$. 2

Find the co-ordinates of A .

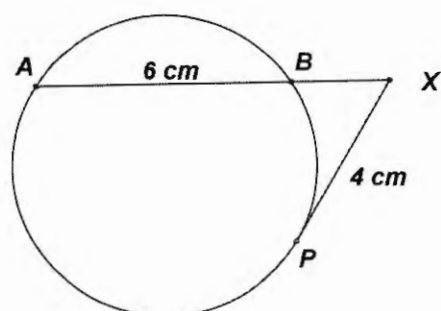
- (c) Use the substitution $t = \tan \frac{x}{2}$ to show that the solutions to the equation 3

$6 + 4\cos x + 5\sin x = 0$, correct to the nearest degree where $0^\circ \leq x \leq 360^\circ$,
are $x = 211^\circ, 252^\circ$.

- (d) Solve for x : 2

$$x + 2\sqrt{x+1} = 7$$

- (e)



The diagram above shows a circle and AB is a chord of length 6 cm .

The tangent to the circle at P meets AB produced at X . $PX = 4\text{ cm}$.

- (i) Find the length of BX . 1

- (ii) Given that AP is a diameter of the circle find the exact length of 2
the radius of the circle.

Question 10 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the domain and range of the function $f(x) = \sqrt{3 - \sqrt{x}}$ 3
- (b) Find the Cartesian equation of the parabola given 1
 $x = t + 3, \quad y = 4 - 2t^2$
- (c) Show that $\tan 2x + \tan x = \frac{\sin 3x}{\cos 2x \cos x}$ 2
- (d) In how many ways can 5 women and 6 men sit around a table.
- (i) Without restrictions. 1
- (ii) If 2 particular men wish to sit next to a particular woman. 2

- (e) From the top of a lighthouse, L , 115 metres above sea level, a container ship, **3**

C , is seen on a bearing of 135°T , at an angle of depression of 10° .

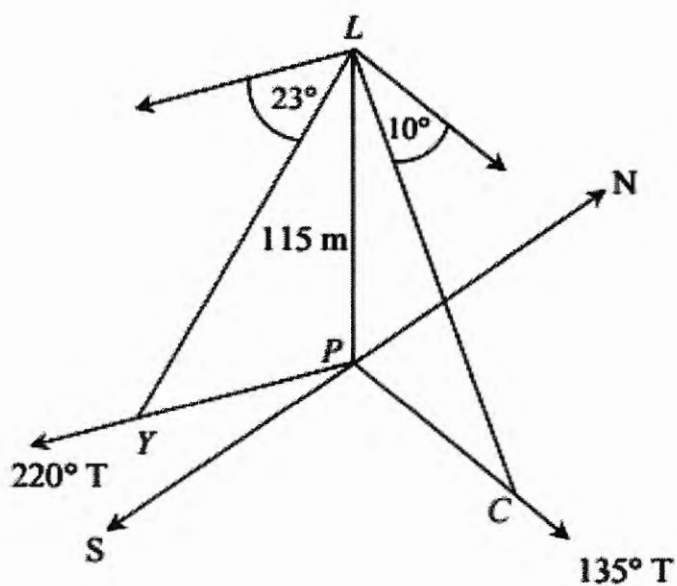
A yacht, Y , is also sighted on a bearing of 220°T , at an angle of depression of 23° .

This is illustrated in the diagram below.

Show that the square of the distance between the two vessels, $(CY)^2$, is given by:

$$(CY)^2 = 115^2 (\tan^2 67^\circ + \tan^2 80^\circ - 2 \tan 67^\circ \tan 80^\circ \cos 85^\circ)$$

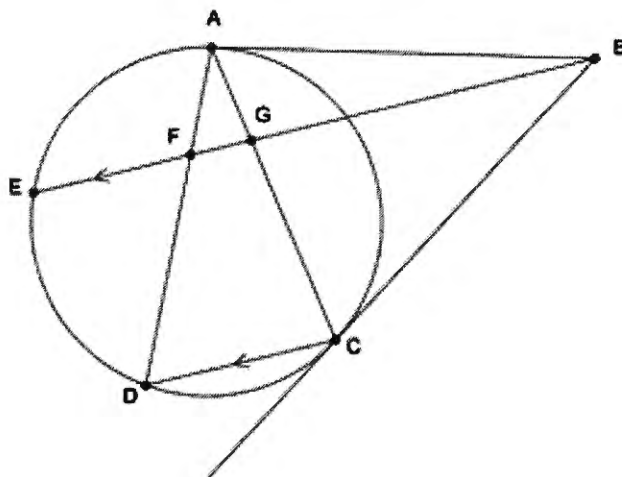
Give clear reasons for each step in your calculations.



Question 11 (12 marks) Use a SEPARATE writing booklet.

(a) Show that $A = \frac{1}{2}$ if $\frac{6^n + 3^n}{2^{n+1} + 2} = A \times 3^n$. 2

(b) In the diagram EB is parallel to DC . Tangents from B meet the circle at A and C .



Copy or trace the diagram into your answer booklet.

(i) Prove that $\angle BCA = \angle BFA$. 2

(ii) Prove $ABCF$ is a cyclic quadrilateral. 1

(c) If α , β and γ are the roots of the equation $2x^3 - 5x^2 - 2x + 6 = 0$, find the values of:

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1

(iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

(iii) $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$ 3

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1 $48 + x + 90 = 180$

$\therefore x = 42$

(B)

2 $\sqrt{3} \sin \theta - \cos \theta$

$R = 2$

$\therefore 2 \sin(\theta - \alpha) = \sqrt{3} \sin \theta - \cos \theta$

$\therefore 2 \cos \alpha = \sqrt{3}$

$2 \sin \alpha = 1$

$\therefore \tan \alpha = \frac{1}{\sqrt{3}}$

$\alpha = 30^\circ$

$\therefore 2 \sin(\theta - 30^\circ)$

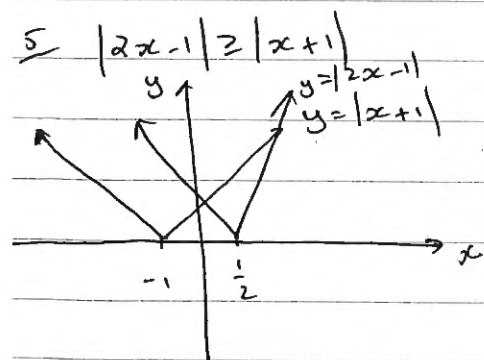
(B)

3 $\frac{6!}{2!}$

(C)

4 $y = \frac{1}{2}(x+1)^3(2-x)$

(B)



pts int: $-(2x-1) = x+1$

$3x = 0 \quad \underline{x = 0}$

$2x-1 = x+1$

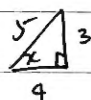
$\underline{x = 2}$

$y = |2x-1|$ above $y = |x+1|$

$x \leq 0, x \geq 2$

(D)

6



Q2

$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

$= \frac{2 \cdot \frac{3}{4}}{1 - \frac{9}{16}}$

$= \frac{-\frac{3}{2}}{\frac{7}{16}}$

$= -\frac{24}{7}$

(B)

7 $P(x) = x^3 - 2x^2 - 5x + 6$

$P(2) \neq 0$

$P(1) = 0$

$P(-3) \neq 0$

$P(-4) \neq 0$

(C)

8

a) $P(x) = (x+3)(x-2)(ax+b)$

$3(-2)(b) = 6$

$\therefore -6b = 6 \quad \therefore b = -1$

$(1)(1)a = 1 \quad \therefore a = 1$

b) $y = 2mx + 1$

$m_1 = 2m$

$y = mx + 7$

$m_2 = m$

$\tan \alpha = \frac{2m - m}{1 + 2m^2}$

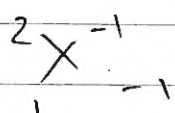
$\therefore \frac{m}{1 + 2m^2} = \frac{1}{3}$

$3m = 1 + 2m^2$

$2m^2 - 3m + 1 = 0$

$(2m-1)(m-1) = 0$

$m = \frac{1}{2}, 1$



$$c) \frac{x+3}{2x} > 1$$

C.V at

$$x=0$$

$$x+3=2x$$

$$x=3$$

$$\begin{array}{c|c|c} x & & x \\ \hline 0 & & 3 \end{array}$$

$$0 \quad 3$$

$$x=-1 \quad \frac{2}{-2} > 1 \quad \times$$

$$x=1 \quad \frac{4}{2} > 1 \quad \checkmark$$

$$x=4 \quad \frac{7}{8} > 1 \quad \times$$

$$0 < x < 3$$

$$d) 6m \quad 5F \quad (5 \text{ chosen } 2F)$$

$$\therefore {}^6C_3 \times {}^5C_2 = 200$$

$$e) \sin 2x = \tan x$$

$$2\sin x \cos x = \frac{\sin x}{\cos x} \quad \cos x \neq 0$$

$$2\sin x \cos^2 x - \sin x = 0$$

$$\sin x (2\cos^2 x - 1) = 0$$

$$\sin x = 0 \quad \left| \quad 2\cos^2 x - 1 = 0 \right.$$

$$\cos^2 x = \frac{1}{2}$$

$$\cos x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = 0, 180^\circ, 45^\circ, 135^\circ$$

Q9

$$a) x^3 - 2x^2 + kx + c = 0$$

$$x = -1, 1, 2$$

$$\Sigma 2p = -1 - 2 + 2$$

$$= -1$$

$$\therefore -1 = \frac{k}{1} \quad \therefore k = -1$$

$$b) A(x, y) \quad B(-2, 1)$$

$$2:3$$

$$3 = \frac{-4+3x}{5}$$

$$4 = \frac{2+3y}{5}$$

$$15 = -4 + 3x$$

$$20 = 2 + 3y$$

$$3x = 19$$

$$3y = 18$$

$$x = \frac{19}{3}$$

$$y = 6$$

$$A\left(\frac{19}{3}, 6\right)$$

$$c) 6 + 4\cos x + 5\sin x = 0$$

$$t = \tan \frac{x}{2}$$

$$\begin{array}{c} 1+t^2 \\ \swarrow \quad \searrow \\ x \quad 2t \\ \downarrow \quad \uparrow \\ 1-t^2 \end{array}$$

$$6 + 4\left(\frac{1-t^2}{1+t^2}\right) + 5\left(\frac{2t}{1+t^2}\right) = 0$$

$$6(1+t^2) + 4(1-t^2) + 10t = 0$$

$$6 + 6t^2 + 4 - 4t^2 + 10t = 0$$

$$2t^2 + 10t + 10 = 0$$

$$t^2 + 5t + 5 = 0$$

$$t = \frac{-5 \pm \sqrt{25-20}}{2}$$

$$= \frac{-5 \pm \sqrt{5}}{2}$$

$$\therefore \tan \frac{x}{2} = \frac{-5 \pm \sqrt{5}}{2} \doteq -1.38, -3.62$$

$$\therefore x = 251^\circ 51', 210^\circ 53'$$

$$d) x + 2\sqrt{x+1} = 7$$

$$2\sqrt{x+1} = 7-x$$

$$4(x+1) = (7-x)^2$$

$$4x+4 = 49-14x+x^2$$

$$x^2 - 18x + 45 = 0$$

$$x=3 \text{ only}$$

$$(x-15)(x-3) = 0$$

solution

$$x = 3, 15$$

$$\text{check } x=3 \quad 3+2\sqrt{4} = 7 \quad \checkmark$$

$$x=15 \quad 15+2\sqrt{16} = 7 \quad \times$$

e) i) $BX \cdot AX = PX^2$

$\therefore x(6+x) = 16$

$6x + x^2 = 16$

$x^2 + 6x - 16 = 0$

$(x+8)(x-2) = 0$

$x = -8, 2 \quad x > 0$

$\therefore x = 2 \quad \text{ie } BX = 2 \text{ cm}$

ii) $AP^2 + PX^2 = AX^2$ (Pythagoras)

($\angle$ between radius & tangent at pt contact = 90°)

$\therefore AP^2 + 16 = 8^2$

$AP^2 = 48$

$\therefore AP = 4\sqrt{3} \text{ cm}$

10

a) $f(x) = \sqrt{3-\sqrt{x}} \quad x \geq 0$

$3-\sqrt{x} \geq 0$

$\sqrt{x} \leq 3$

$x \leq 9$

domain: $0 \leq x \leq 9$

range: $0 \leq y \leq \sqrt{3}$

max $\sqrt{3}$

min 0

b) $x = t+3 \Rightarrow t = x-3$

$y = 4 - 2t^2$

$y = 4 - 2(x-3)^2$

$y = 4 - 2(x^2 - 6x + 9)$

$y = 4 - 2x^2 + 12x - 18$

$y = -2x^2 + 12x - 14$

c) $\tan 2x + \tan x = \frac{\sin 3x}{\cos 2x \cos x}$

LHS = $\frac{\sin 2x}{\cos 2x} + \frac{\sin x}{\cos x}$

= $\frac{\sin 2x \cos x + \sin x \cos 2x}{\cos x \cos 2x}$

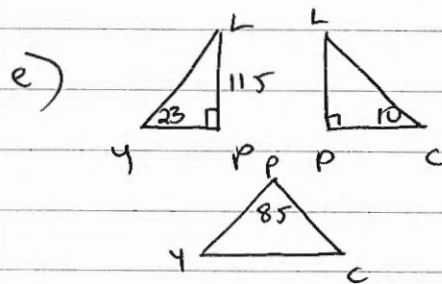
= $\frac{\sin(2x+x)}{\cos x \cos 2x}$

= $\frac{\sin 3x}{\cos x \cos 2x}$

= RHS //

d) i) $10! = 3628800$

ii) $8! \times 2 = 80640$



$\tan 23^\circ = \frac{115}{PY} \quad \tan 10^\circ = \frac{115}{PC}$

$PY = \frac{115}{\tan 23^\circ}$

$PC = \frac{115}{\tan 10^\circ}$

$\angle YPC = 220 - 135 = 85^\circ$

$\therefore CY^2 = PY^2 + PC^2 - 2 \cdot PY \cdot PC \cdot \cos 85^\circ$

(Cosine Rule)

$\therefore CY^2 = \frac{115^2}{\tan^2 23^\circ} + \frac{115^2}{\tan^2 10^\circ} - \frac{2 \cdot 115 \cdot 115}{\tan 23^\circ \tan 10^\circ}$

$CY^2 = 115^2 \left(\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 10^\circ} - \frac{2 \cos 85^\circ}{\tan 23^\circ \tan 10^\circ} \right)$

= $115^2 (\cot^2 23^\circ + \cot^2 10^\circ - 2 \cot 23^\circ \cot 10^\circ \cos 85^\circ)$

= $115^2 (\tan^2 67^\circ + \tan^2 80^\circ - 2 \tan 67^\circ \tan 80^\circ \cos 85^\circ)$

$\cot \theta = \tan(90^\circ - \theta)$

$\cos 85^\circ$

Q11

$$a) \frac{6^n + 3^n}{2^{n+1} + 2} = A \times 3^n$$

$$\begin{aligned} \text{LHS} &= \frac{(2 \times 3)^n + 3^n}{2 \cdot 2^n + 2} \\ &= \frac{2^n \cdot 3^n + 3^n}{2 \cdot 2^n + 2} \\ &= \frac{3^n(2^n + 1)}{2(2^n + 1)} \end{aligned}$$

$$= \frac{3^n}{2} = \frac{1}{2} \times 3^n$$

$$\therefore A = \frac{1}{2}$$

b) i) $\angle BCA = \angle CDA$

($\angle$ between tangent & chord at pt of contact = $\angle$ in alternate segment)

$$\angle CDA = \angle BFA$$

(Corresponding $\angle$ s =, $EB \parallel CD$)

ii) $ABCF$ is cyclic quad

$$\angle BCA = \angle BFA \text{ (from above)}$$

$$\therefore ABCF \text{ is cyclic quad}$$

(= $\angle$'s at circumference from same arc)

c) $2x^3 - 5x^2 - 2x + 6 = 0$

i) $\alpha + \beta + \gamma = \frac{5}{2}$

ii) $\alpha\beta + \beta\gamma + \alpha\gamma = \frac{-2}{2} = -1$

iii) $\alpha^2 + \beta^2 + \gamma^2$:

$$\begin{aligned} (\alpha + \beta + \gamma)^2 &= \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma \\ &\quad + \alpha\gamma + \beta\gamma + \gamma^2 \\ &= \alpha^2 + \beta^2 + \gamma^2 + 2(\sum \alpha\beta) \quad (1) \end{aligned}$$

$$\begin{aligned} \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\sum \alpha)^2 - 2(\sum \alpha\beta) \\ &= \left(\frac{5}{2}\right)^2 - 2(-1) \quad (1) \end{aligned}$$

$$\begin{aligned} &= \frac{25}{4} + 2 \\ &= \frac{33}{4} \end{aligned}$$

iii) $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$

$$= \frac{\gamma^2 + \alpha^2 + \beta^2}{\alpha^2\beta^2\gamma^2} \quad (1)$$

$$= \frac{\frac{33}{4}}{(-3)^2}$$

$$\alpha\beta\gamma = \frac{-6}{2} = -3 \quad (1)$$

$$= \frac{11}{12} \quad (1)$$